Jaime Polanco-Jimenez

Ph.D. in Economics, Pontificia Universidad Javeriana & KU Leuven

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Research Interests

Applied Microeconomics, Human Capital Formation, Economics of Artificial Intelligence, Labor Economics, and Development Economics.

Education

KU Leuven — **Aug 2020 – March 2026**
Ph.D. in Economics (Joint Degree) — *Leuven, Belgium*

Pontificia Universidad Javeriana — **Aug 2020 – May 2026**
Ph.D. in Economics (Joint Degree) — *Bogota, Colombia*

Pontificia Universidad Javeriana — **Nov 2024**
M.S. in Economics — *Bogota, Colombia*

Udmurt State University — **Sep 2018 – Jul 2020**
M.S. in Economics — *Izhevsk, Russia*

Udmurt State University — **Sep 2017 – Jul 2019**
M.S. in Oil and Gas Engineering — *Izhevsk, Russia*

University of America — **2016**
Bachelor in Oil and Gas Engineering — *Bogota, Colombia*

Publications

- **(Job Market Paper)** Polanco-Jimenez, J. & De Witte, K. (2025). *The Tailoring Premium: AI Design, Behavioral Engagement, and Human Capital Formation.* Accepted at *Journal of Economic Behavior and Organization*. [Tracking Link]
- Polanco-Jimenez, J. (2025). *Road to the Future: Identifying Impacts of Roads on Education in Colombia. International Journal of Educational Development*, 114, 103173.
- **(Policy Report)** Polanco-Jimenez, J. & De Witte, K. (2026). *Tailored AI can help address teacher shortages.* Zenodo. [DOI: 10.5281/zenodo.20393189]
- **(Policy Report)** Polanco-Jimenez, J. & De Witte, K. (2025). *Factors Influencing STEM Participation and Effective Intervention Strategies.* European Commission, Directorate-General for Education, Youth, Sport and Culture / NESET Report, Publications Office of the European Union. [EU Publications] [DOI]
- **(Book Chapter)** Castañeda, B. & Polanco-Jimenez, J. (2026). Decisiones educativas de los jóvenes en tiempos de transición. In A. Álvarez, J. Guerra-España, & M. Hofstetter (Eds.), *Retos económicos de la transición energética en Colombia* (pp. 125–142). Ediciones Uniandes. [DOI] [Book Link]

Working Papers and Books

- Polanco-Jimenez, J., De Witte, K., & Bernal, G. L. (2025). *Does Coeducation Close the STEM Gap? Evidence from a Single-Sex Schooling Transition in Colombia.* Under review at *Journal of Development Economics.*
- Lozano, J. & Polanco-Jimenez, J. (2025). *Financial Literacy and Business Success: Evidence from an RCT in Colombia.*
- Polanco-Jimenez, J., & Posso, C. (2025). *Black Gold and Dull Minds: The Impact of Oil Exploration Announcements on Education in Colombia.*
- Polanco-Jimenez, J., De Witte, K., & François, M. (2026). *Heterogeneous Effects of Persona Design on the Educational Aspirations.*
- Polanco-Jimenez, J., Sevilla, A., & Vicente, I. (2026). *Complementarity Between Expertise and Tasks: Evidence from a Purpose-Built AI System.*
- Polanco-Jimenez, J. & Montero-Mestre, J. L. (2026). *Drilling into the Forest: How Exploration Announcements Reshape Land and Crops in Colombia.*
- **(Book)** Polanco-Jimenez, J. & De Witte, K. (2026). *The Algorithmic Ladder: RCTs, Artificial Intelligence, and the Future of Human Capital — A Comprehensive Framework for Policy-Relevant Science.* Under review, Leuven University Press.

Academic and Research Experience

Pontificia Universidad Javeriana — **Jul 2026 – Present**
Professor of Statistics — *Bogota, Colombia*

London School of Economics and Political Science — **Since March 2026**
Visiting Fellow at WISPPRH — *London, UK*

KU Leuven – LEER, Belgium — **Since March 2026**
Associate Researcher — *Leuven, Belgium*

Pontificia Universidad Javeriana — **Oct 2022 – 2024**
Graduate Teaching Assistant – Advanced Econometrics I & II — *Instructor: Adrian Garlati*

Banco de la República, Colombia — **Aug 2022 – Mar 2023**
Research Assistant — *Joint research on macroeconomic shocks and human capital.*

Pontificia Universidad Javeriana — **Oct 2020 – Aug 2021**
Research Assistant — *Laboratorio Economía de la Educación (LEE)*

Bogotá Summer School in Economics — **2021**
Graduate Teaching Assistant — *Real Analysis (Instructor: Andres Carvajal)*

Extension and Professional Services

Servinformación — **Oct 2022 - Current**
Consultant – Senior Data Scientist — *Bogotá, Colombia*

Fiscal Observatory, Universidad Javeriana — **Aug 2023 – Dec 2023**
Consultant — *Policy and fiscal analyses.*

Secretaría de Salud de Bogotá — **Oct 2020 – Aug 2022**
Data Scientist — *Analytical support during public health operations.*

RCT Registrations + Ongoing Projects

- Polanco-Jimenez, J. & Sevilla, A. (2026). "Evaluating an AI-Powered Research Development Tool for Academic Productivity and Well-being." *AEA RCT Registry*. January 28. DOI
- Polanco-Jimenez, J. & De Witte, K. (2025). "A Randomized Controlled Trial of an AI-Based Guidance Tool for Post-Secondary Planning." *AEA RCT Registry*. DOI
- Polanco-Jimenez, J. & De Witte, K. (2024). "The Tailoring Premium: How AI Design Unlocks Student Engagement and Learning." *AEA RCT Registry*. DOI

Scholarships and Grants

- Google Cloud Platform Research Credits Program, 2024–2026. Awarded $5,000 USD.
- Doctoral Fellowship, Pontificia Universidad Javeriana, 2020. Merit-based full scholarship.
- Russian Government Scholarship, Udmurt State University, 2018. International grant.

Table 1: Technical Skills and Professional Certifications

Category	Details
Languages	Spanish (Native), English (Fluent), Russian (Fluent)
Programming	Python, R, Stata, SQL, LaTeX (github.com/polanco-jaime)
Methodologies	Causal Inference (RCTs, DiD, RDD), Applied Econometrics, Machine Learning, Cloud Computing (GCP)
Certifications	Google Cloud Professional Machine Learning Engineer; Google Cloud Professional Data Engineer

Notes: Certifications verified by Google Cloud Professional program. Methodological expertise focuses on causal identification in human capital research.

References

Kristof De Witte
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Vice-Dean for Research and Impact
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Almudena Sevilla, CBE
Professor of Economic & Social Policy
London School of Economics (LSE)
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Gloria Bernal
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